

EE2211 Introduction to Machine Learning

Lecture 1

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**Office Hour: Wed 9:30 – 10:30 AM, or by email
(Week 2-4, Week 10-12)**

Course Contents

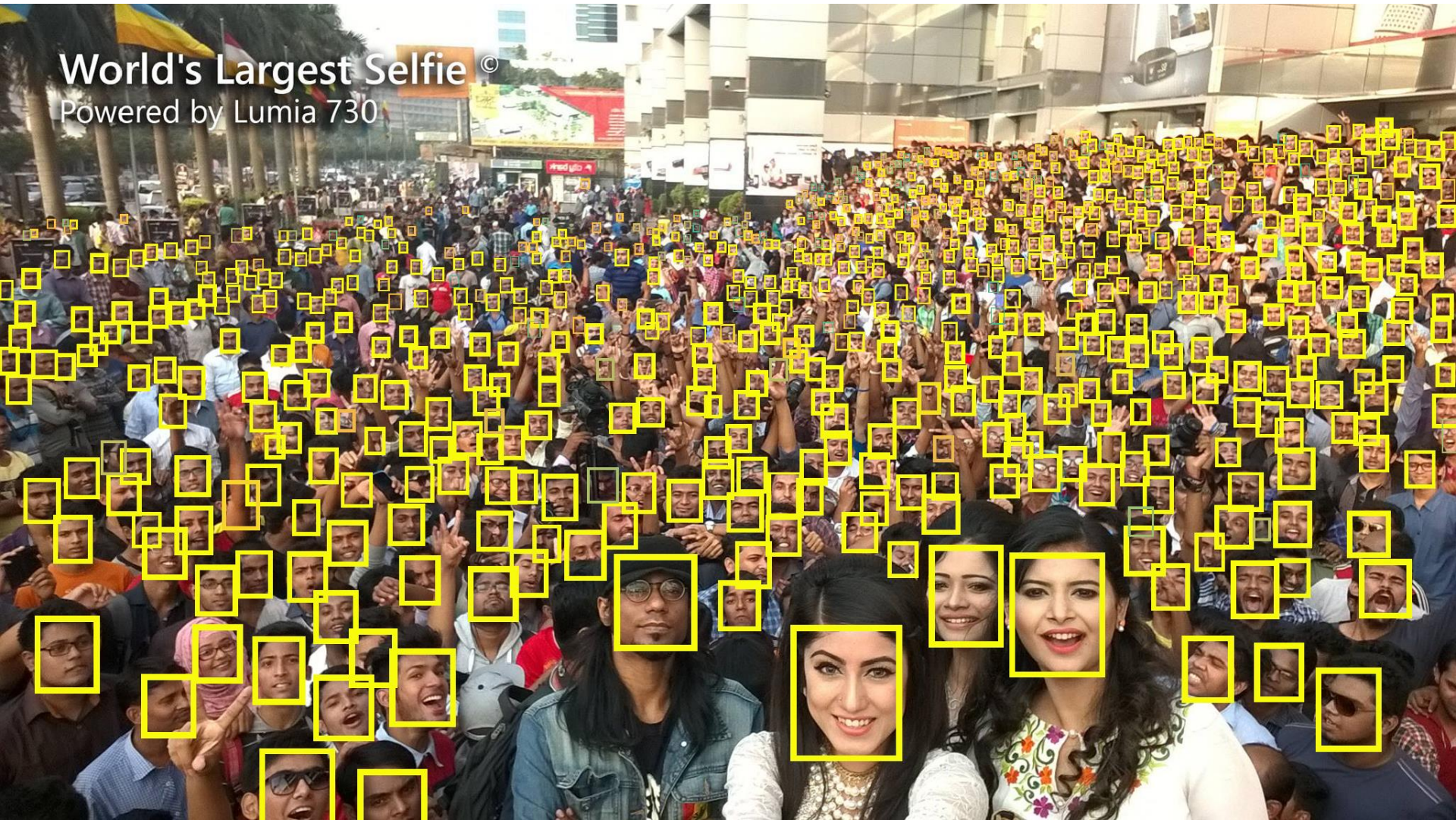
- Introduction and Preliminaries (Xinchao)
 - Introduction
 - Data Engineering
 - Introduction to Probability and Statistics
- Fundamental Machine Learning Algorithms I (Helen)
 - Systems of linear equations
 - Least squares, Linear regression
 - Ridge regression, Polynomial regression
- Fundamental Machine Learning Algorithms II (Helen)
 - Over-fitting, bias/variance trade-off
 - Optimization, Gradient descent
 - Decision Trees, Random Forest
- Performance and More Algorithms (Xinchao)
 - Performance Issues
 - K-means Clustering
 - Neural Networks

World's Largest Selfie

World's Largest Selfie ©
Powered by Lumia 730

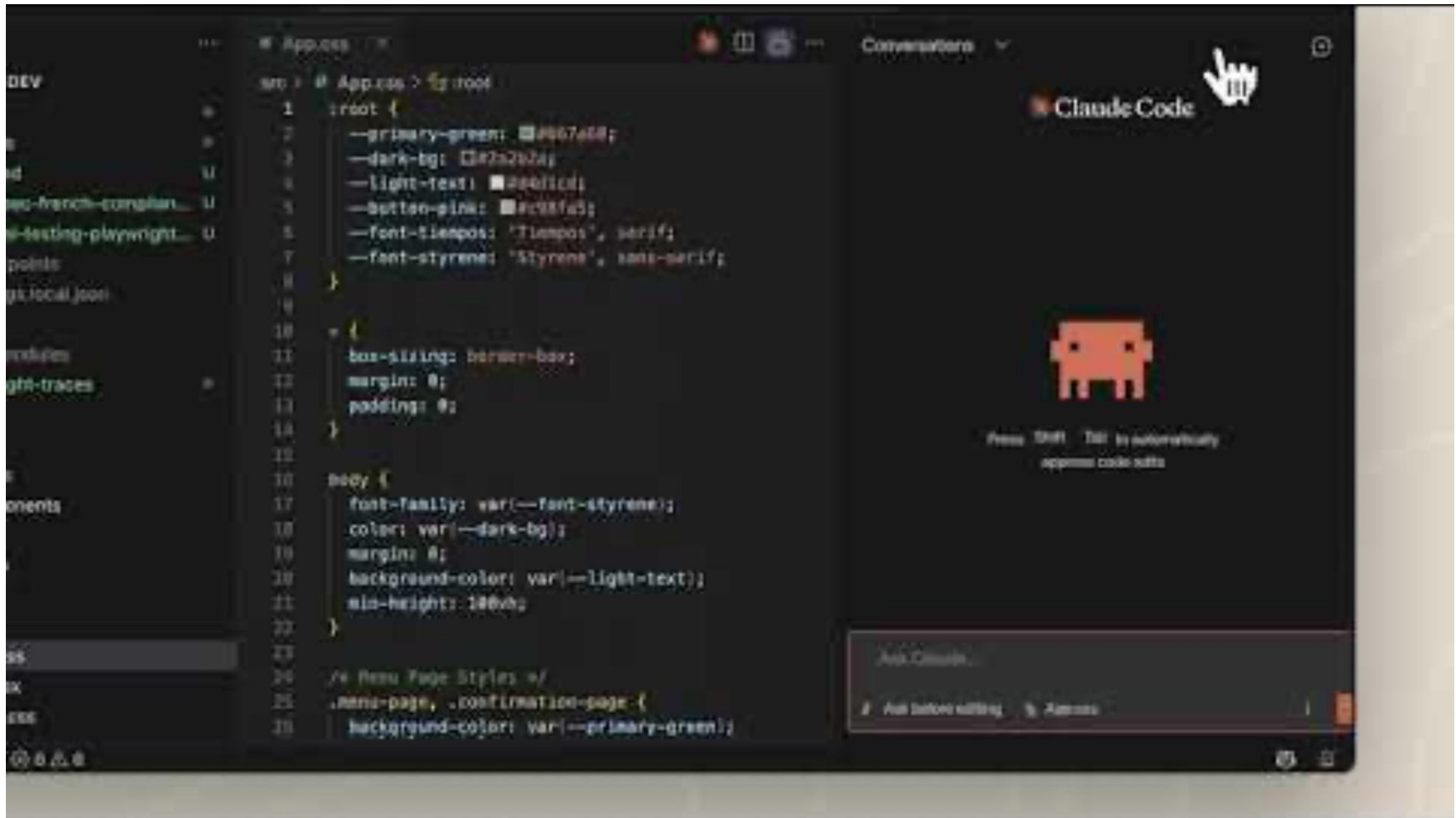


World's Largest Selfie



Recent Advances: Generative AI (GenAI)

Claude Code



<https://www.anthropic.com/news/enabling-claude-code-to-work-more-autonomously>

Outline

- What is machine learning?
 - Three Definition(s)
- When do we need machine learning?
 - Sometimes we need, sometimes we don't
- Applications of machine learning
- Types of machine learning
 - Supervised, Unsupervised, Reinforcement Learning
- Walking through a toy example on classification
- Inductive vs. Deductive Reasoning

What is machine learning?

Learning is any process by which a system improves performance from experience.

- Herbert Simon

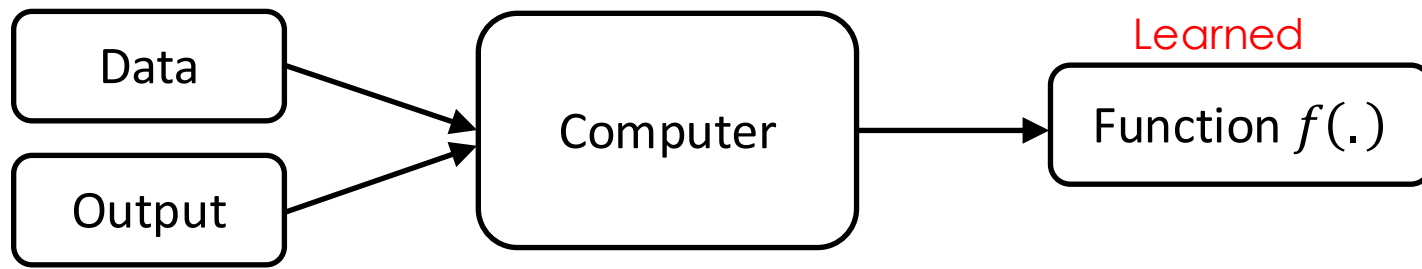
A computer program is said to learn

- from **experience E**
- with respect to some class of **tasks T**
- and **performance measure P** ,

if its performance at tasks in T , as measured by P , improves with experience E .

- Tom Mitchell

Machine Learning (Supervised Learning)



Data

Output



Cat

$$f(\text{cat image}) = \text{'cat'}$$

⋮

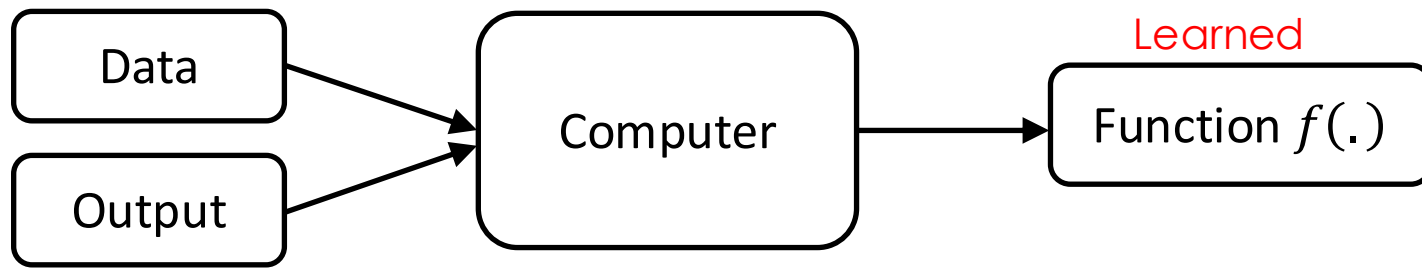
→ $f(.)$ such that



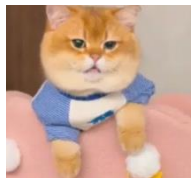
Dog

$$f(\text{dog image}) = \text{'dog'}$$

Machine Learning (Supervised Learning)



Data **Output**



Cat

⋮

→ $f(.)$



Dog

When applied

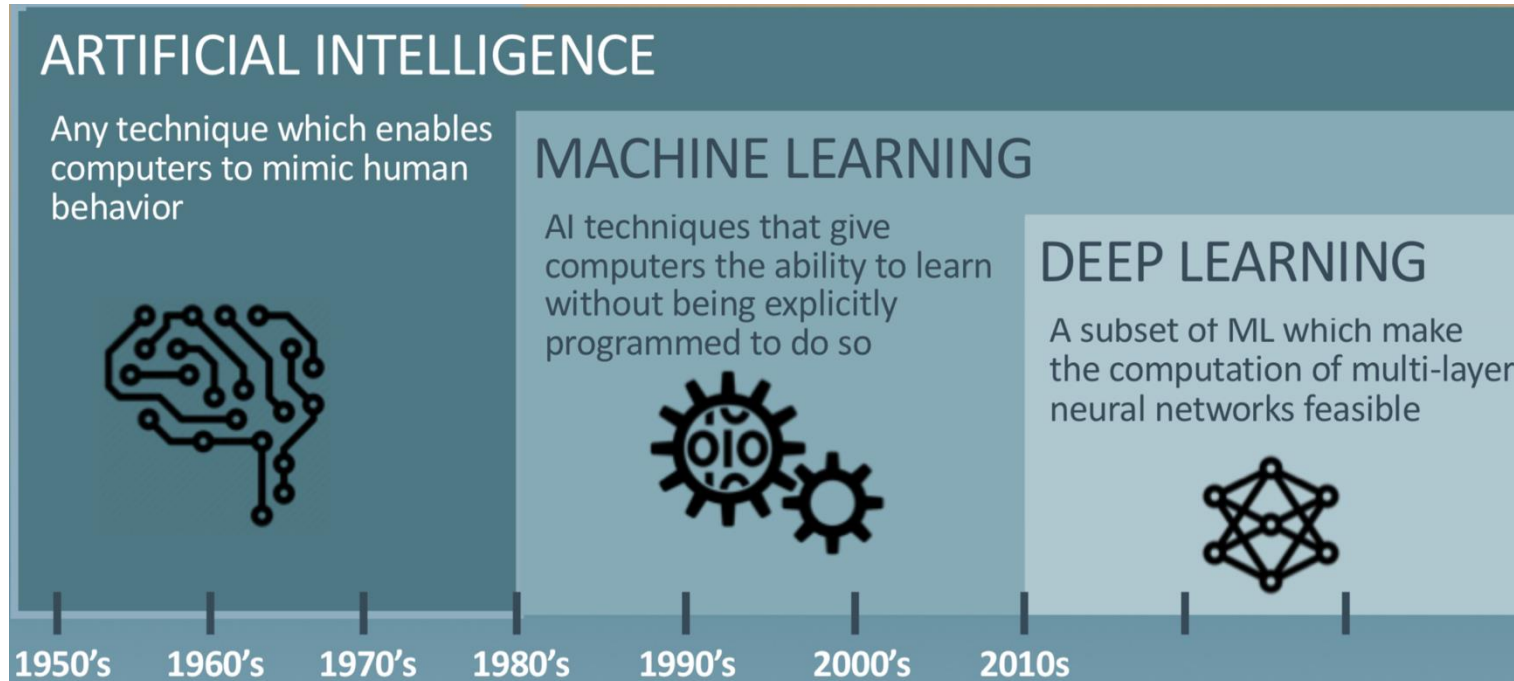
$f($  $) \rightarrow$ **Cat !**

New image

Machine Learning: field of study that gives computers the ability to learn without being explicitly programmed

- Arthur Samuel

AI, Machine Learning, and Deep Learning



Example of AI but not ML: Deductive Reasoning

NUS is in Singapore, Singapore is in Asia $\rightarrow$ NUS is in Asia

What about GenAI?

When do we need machine learning?

Lack of human expertise
(Navigating on Mars)



Involves huge amount of data
(Genomics)



Learning is not always useful:

No need to “learn” to calculate payroll!

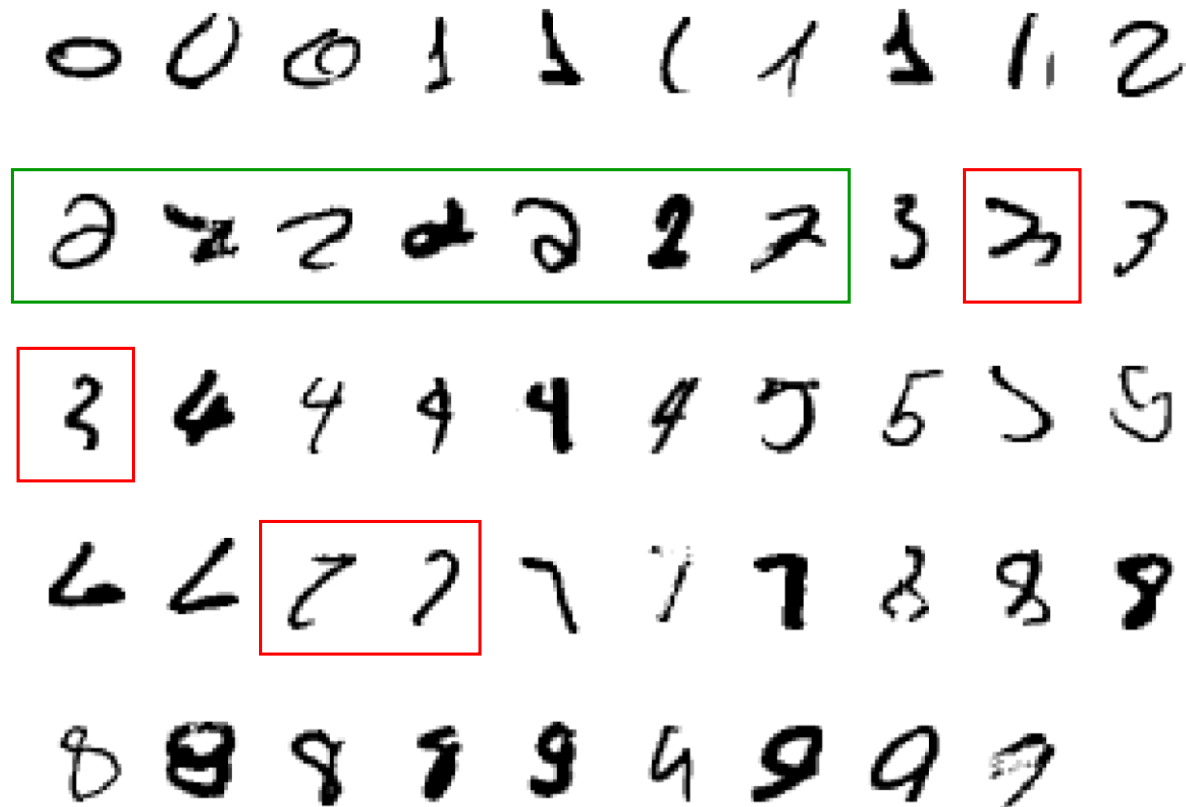
$\text{My Salary} = \text{Days_of_work} * \text{Daily Salary} + \text{Bonus}$

Application of Machine Learning

Task T, Performance P, Experience E

T: Digit Recognition
P: Classification Accuracy
E: Labelled Images

4 “four”
3 “three”



Labels -> Supervision!

Application of Machine Learning

Task T, Performance P, Experience E

T: Email Categorization

P: Classification Accuracy

E: Email Data, Some Labelled



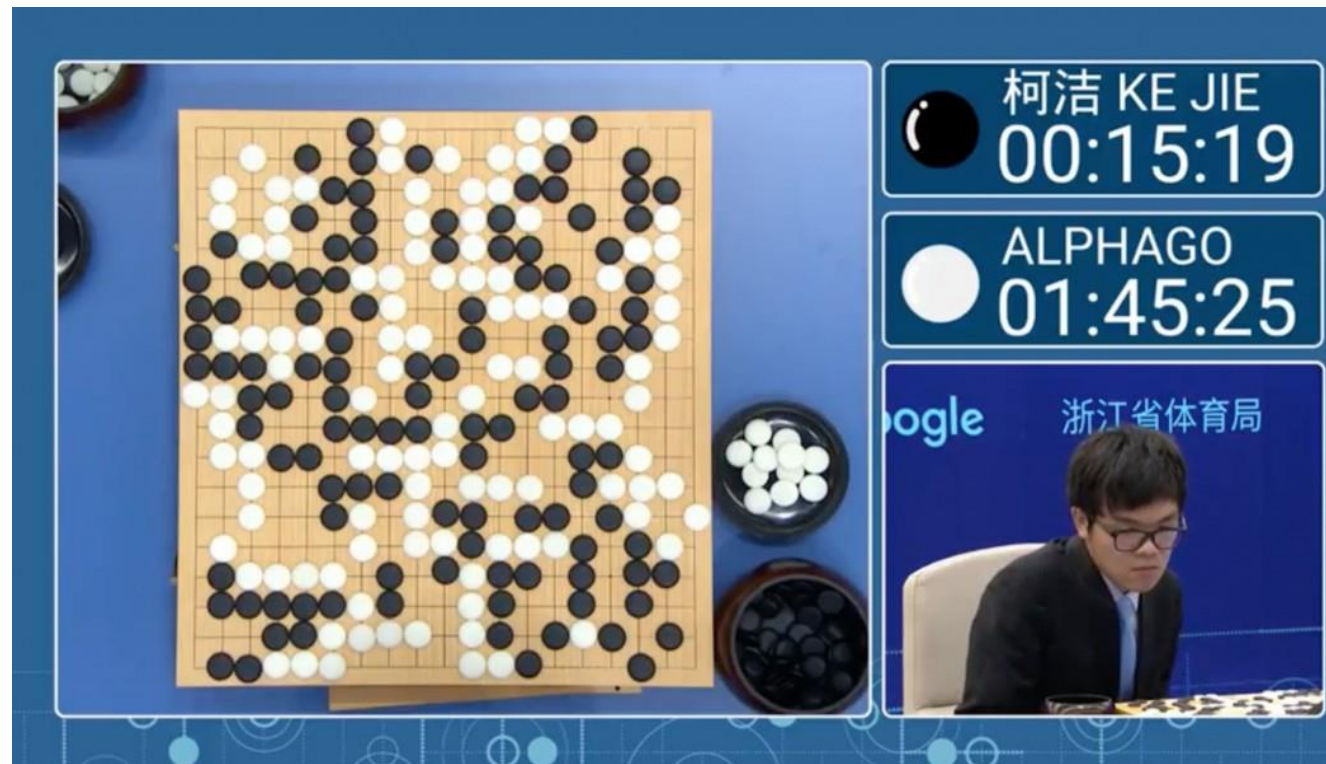
Application of Machine Learning

Task T, Performance P, Experience E

T: Playing Go Game

P: Chances of Winning

E: Records of Past Games



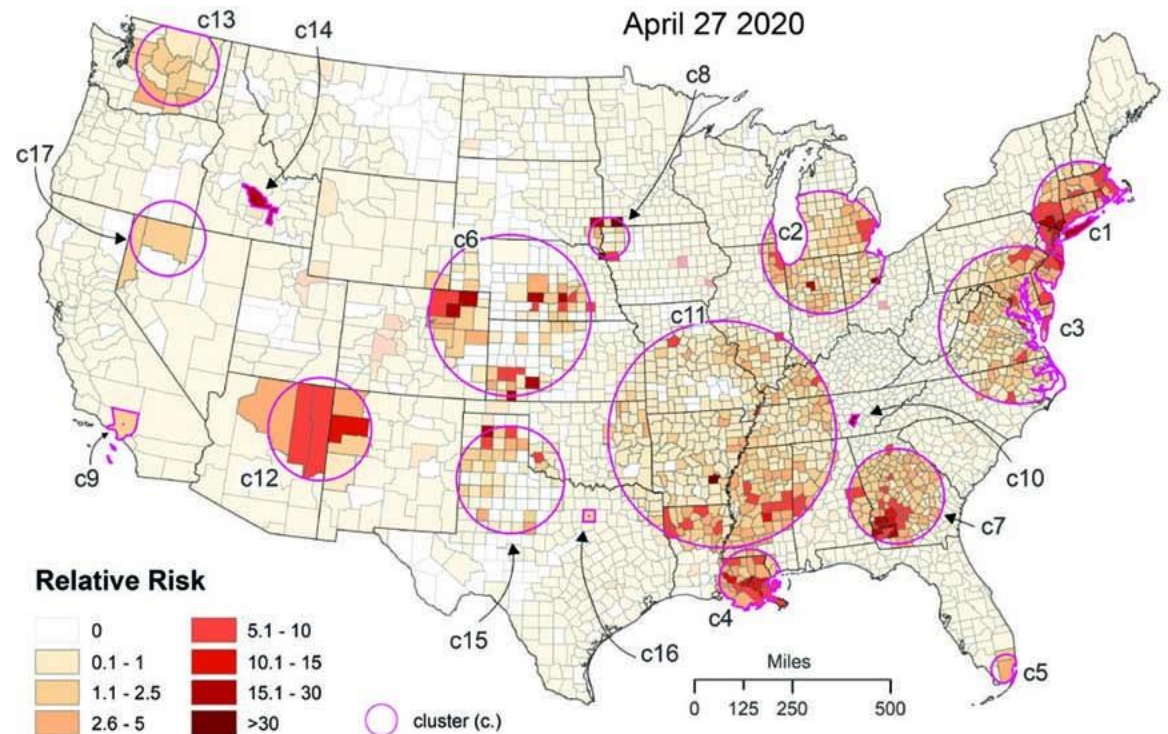
Application of Machine Learning

Task T, Performance P, Experience E

T: Identifying Covid-19 Clusters

P: Small Internal Distances
Larger External Distances

E: Records of Patients





Web Search Engine



Product Recommendation



Language Translation

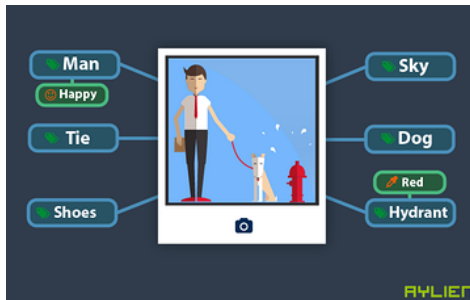


Photo Tagging



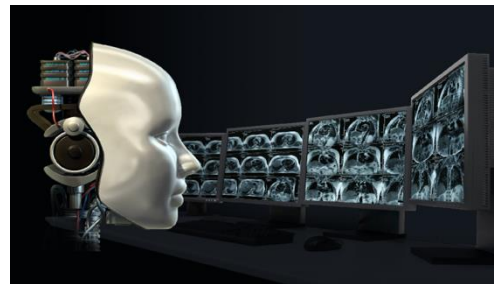
Virtual Personal Assistant



Portfolio Management



Traffic Prediction



Medical Diagnosis



Algorithmic Trading

Types of Machine Learning

Supervised Learning

Input:

- 1) Training Samples,
- 2) Desired Output (Teacher/Supervision)

Output:

A rule that maps input to output

Unsupervised Learning

Input:

Samples

Output:

Underlying patterns in data

Reinforcement Learning

Input:

Sequence of States, Actions, and Delayed Rewards

Output:

Action Strategy: a rule that maps the environment to action

Types of Machine Learning

Supervised Learning

Input:

- 1) Training Samples,
- 2) Desired Output (Teacher/Supervision)

Output:

A rule that maps input to output

Data Output



Cat

⋮

→ $f(\cdot)$ such that



Dog

Unsupervised Learning

Input:

Samples

Output:

Underlying patterns in data

Reinforcement Learning

Input:

Sequence of States, Actions, and Delayed Rewards

Output:

Action Strategy: a rule that maps the environment to action

$$f(\text{cat image}) = \text{'cat'}$$

$$f(\text{dog image}) = \text{'dog'}$$

Supervised Learning

Data

x



Regression



y (Continuous)

x



Classification

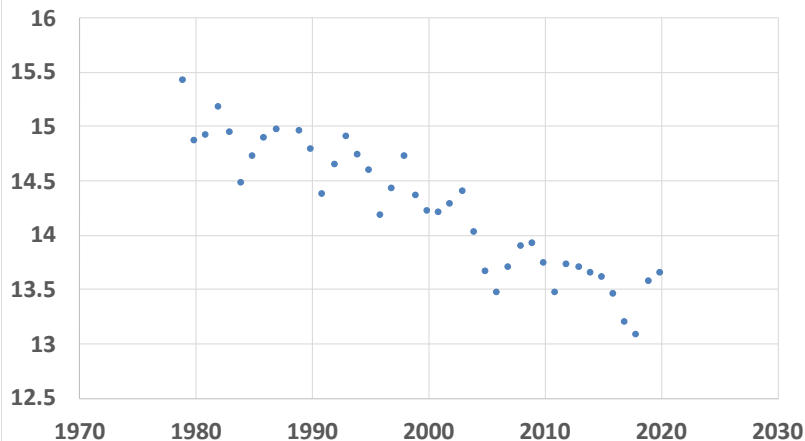


y (Categorical)

Regression

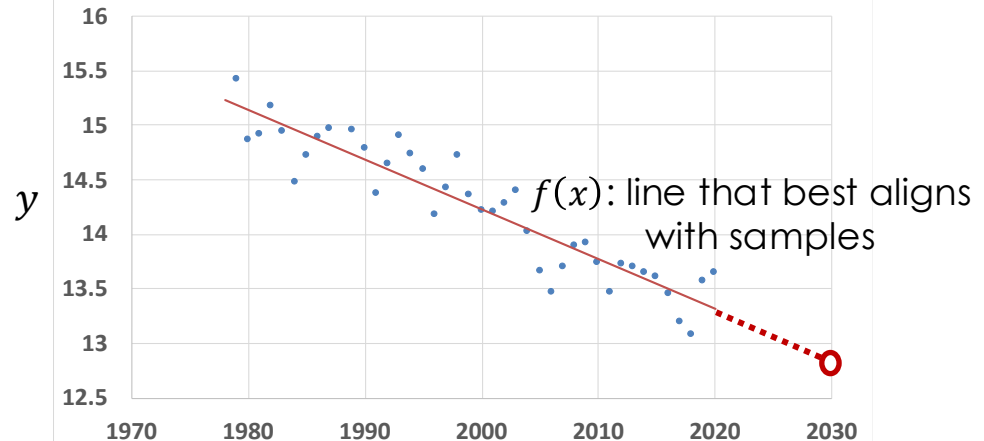
- Given $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_N, y_N)$
- Learn a function $f(\mathbf{x})$ to predict real-valued y given $\mathbf{x}$

Arctic Sea Ice Extent in January (in million sq km)



x

Arctic Sea Ice Extent in January (in million sq km)



x

Supervised Learning

Data

x



Regression



Output

y (Continuous)

x



Classification

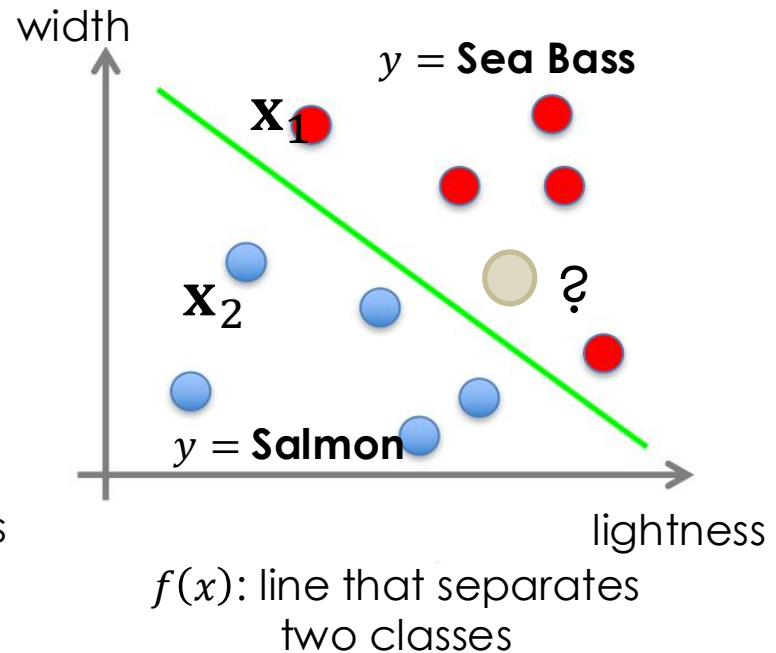
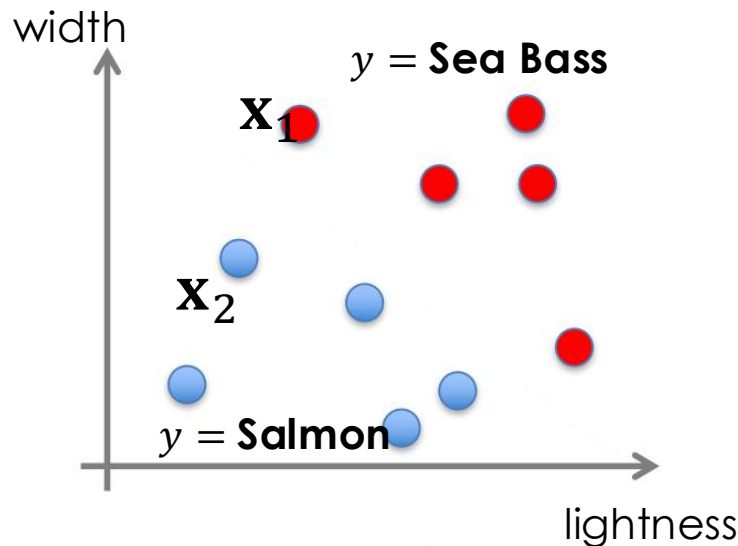


y (Categorical)

Classification

- Given $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_N, y_N)$
- Learn a function $f(\mathbf{x})$ to predict categorical y given $\mathbf{x}$

Feature Space



Types of Machine Learning

Supervised Learning

Input:

- 1) Training Samples,
- 2) Desired Output (Teacher/Supervision)

Output:

A rule that maps input to output

Unsupervised Learning

Input:

Samples

Output:

Underlying patterns in data

Reinforcement Learning

Input:

Sequence of States, Actions, and Delayed Rewards

Output:

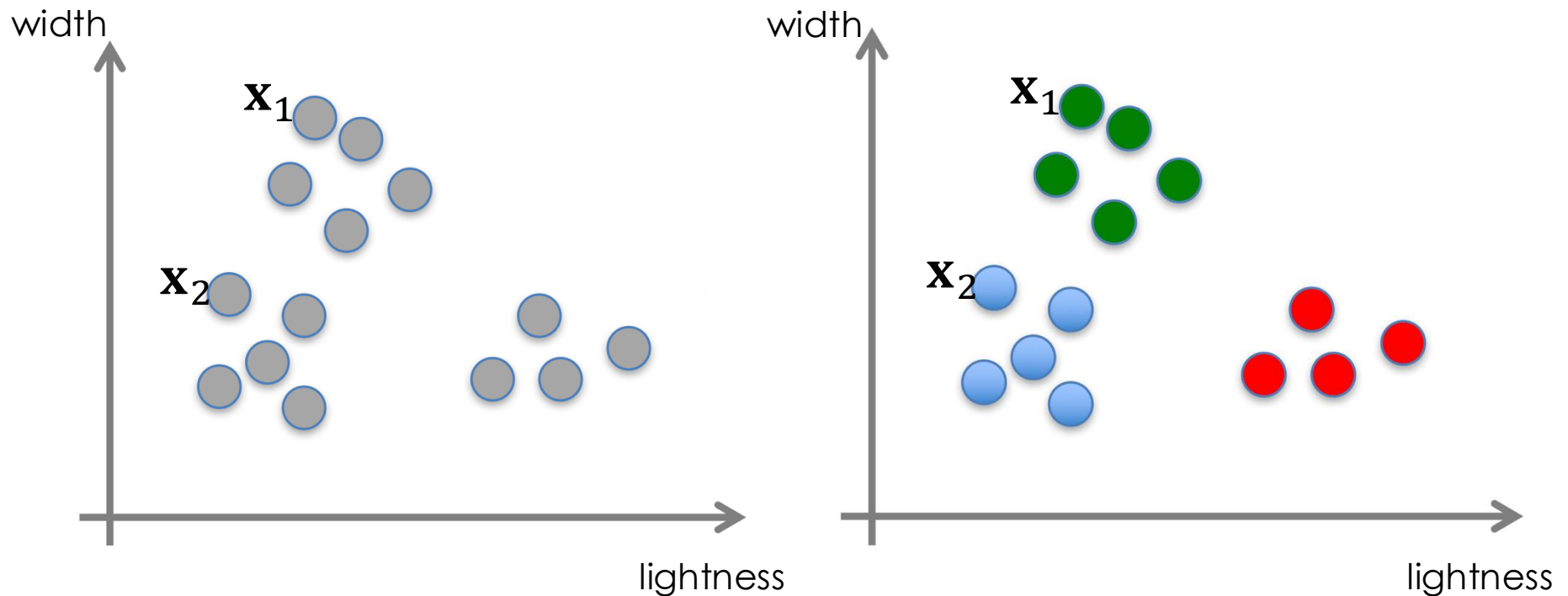
Action Strategy: a rule that maps the environment to action

**Key different w.r.t. supervised learning:
No Label/Supervision is given!**

Unsupervised Learning

Clustering

- Given $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N$, without labels
- Output Hidden Structure Behind



No Label/Supervision is given!

Types of Machine Learning

Supervised Learning

Input:

- 1) Training Samples,
- 2) Desired Output (Teacher/Supervision)

Output:

A rule that maps input to output

Unsupervised Learning

Input:

Samples

Output:

Underlying patterns in data

Reinforcement Learning

Input:

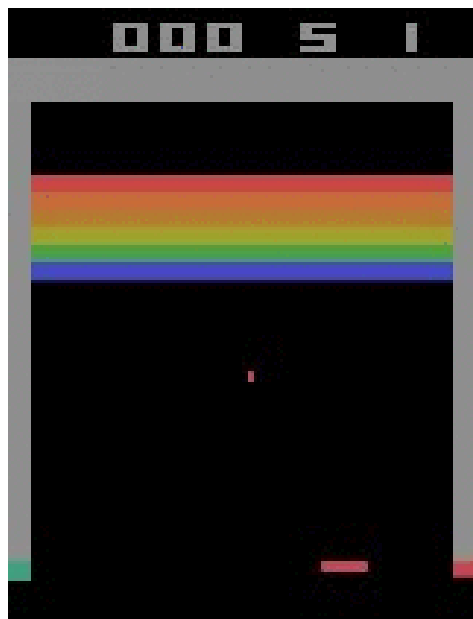
Sequence of States, Actions, and Delayed Rewards

Output:

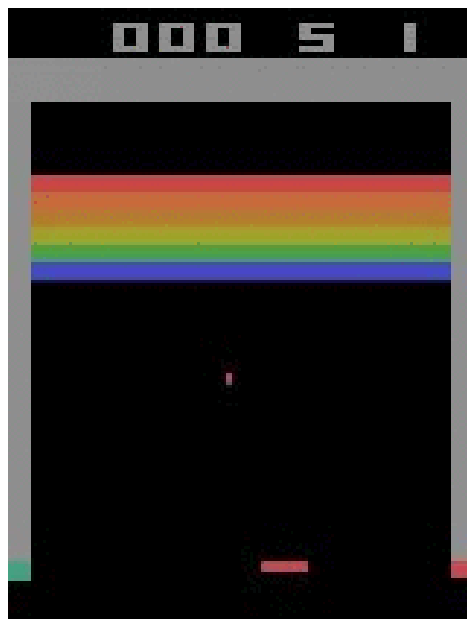
Action Strategy: a rule that maps the environment to action

Reinforcement Learning

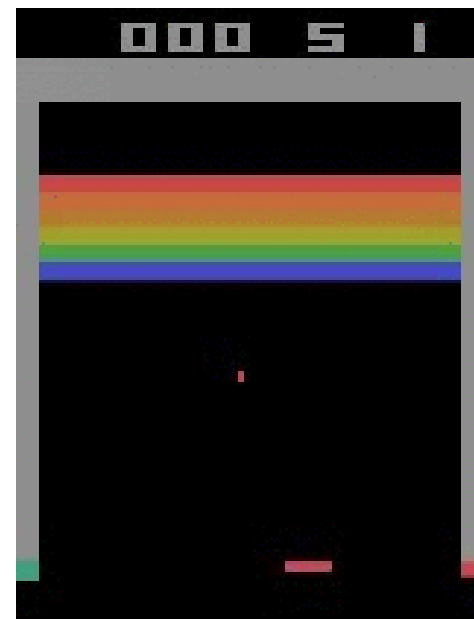
Breakout Game



Initial Performance



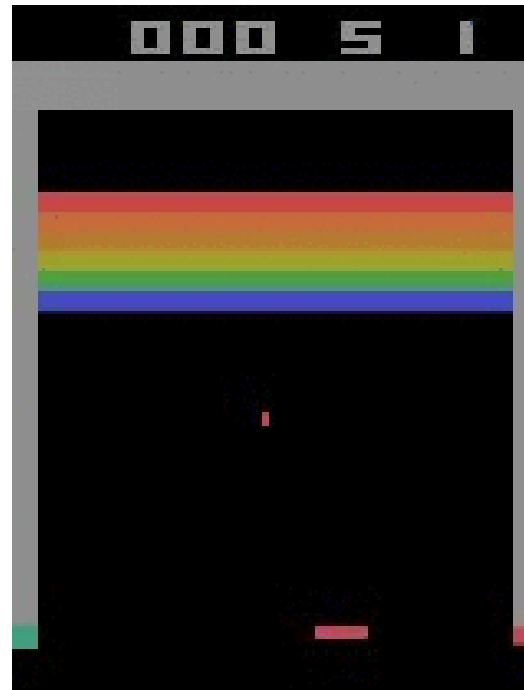
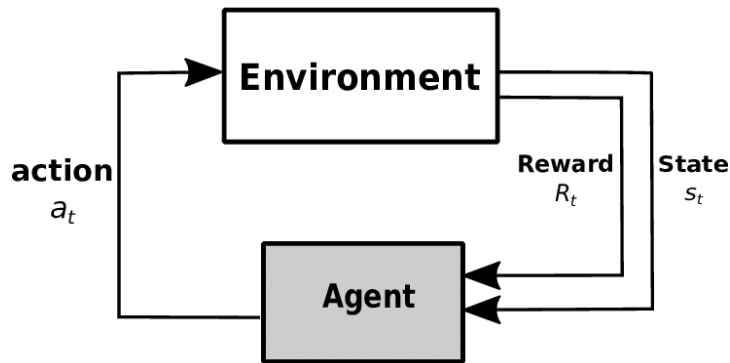
Training 15 minutes



Training 30 minutes

Reinforcement Learning

- Given sequence of states $\mathcal{S}$ and actions $\mathcal{A}$ with (delayed) rewards $\mathcal{R}$
- Output a policy $\pi(a, s)$, to guide us what action a to take in state s



$\mathcal{S}$: Ball Location,
Paddle Location, Bricks

$\mathcal{A}$: left, right

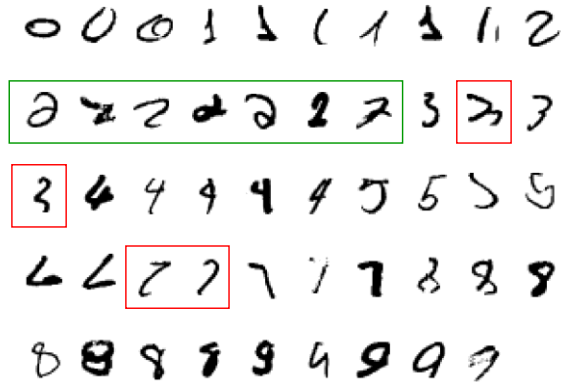
$\mathcal{R}$:
positive reward
Knocking a brick,
clearing all bricks

negative reward
Missing the ball

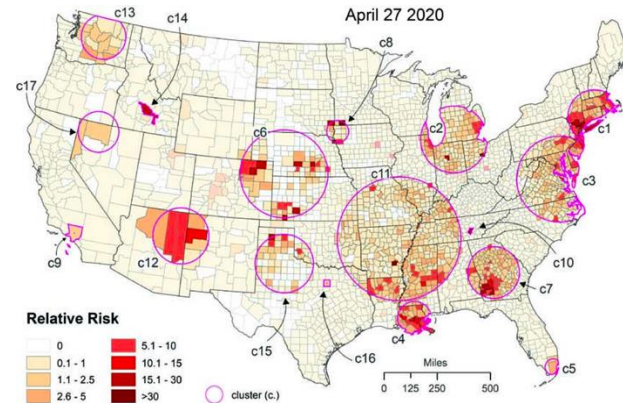
zero reward
Cases in between

Supervised
Unsupervised
Reinforcement

Quiz Time!



Supervised



Unsupervised

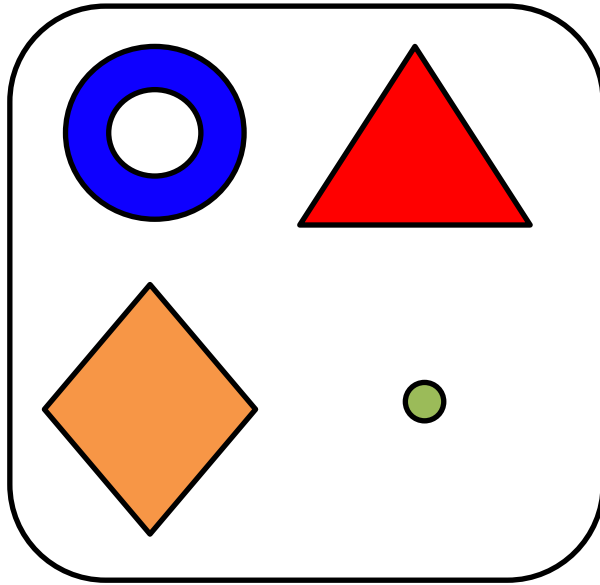


Supervised

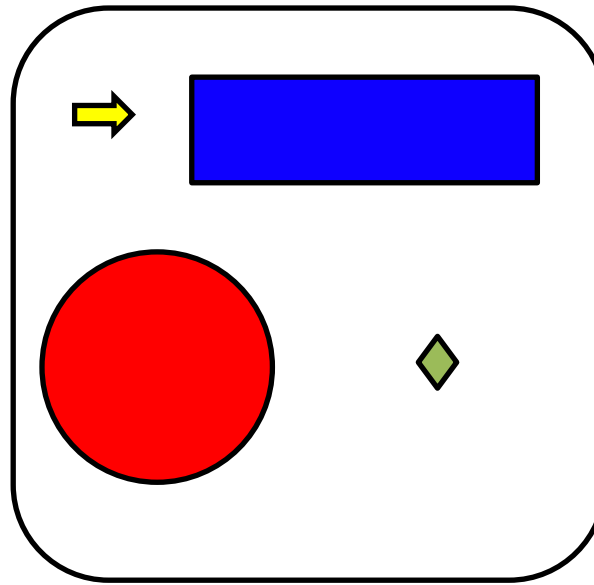


Reinforcement

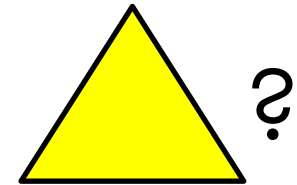
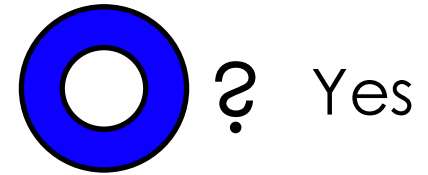
Walking Through A Toy Example: Classifying Tokens



Yes



No

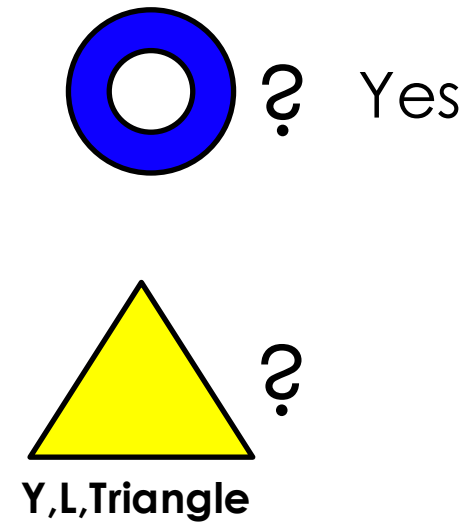
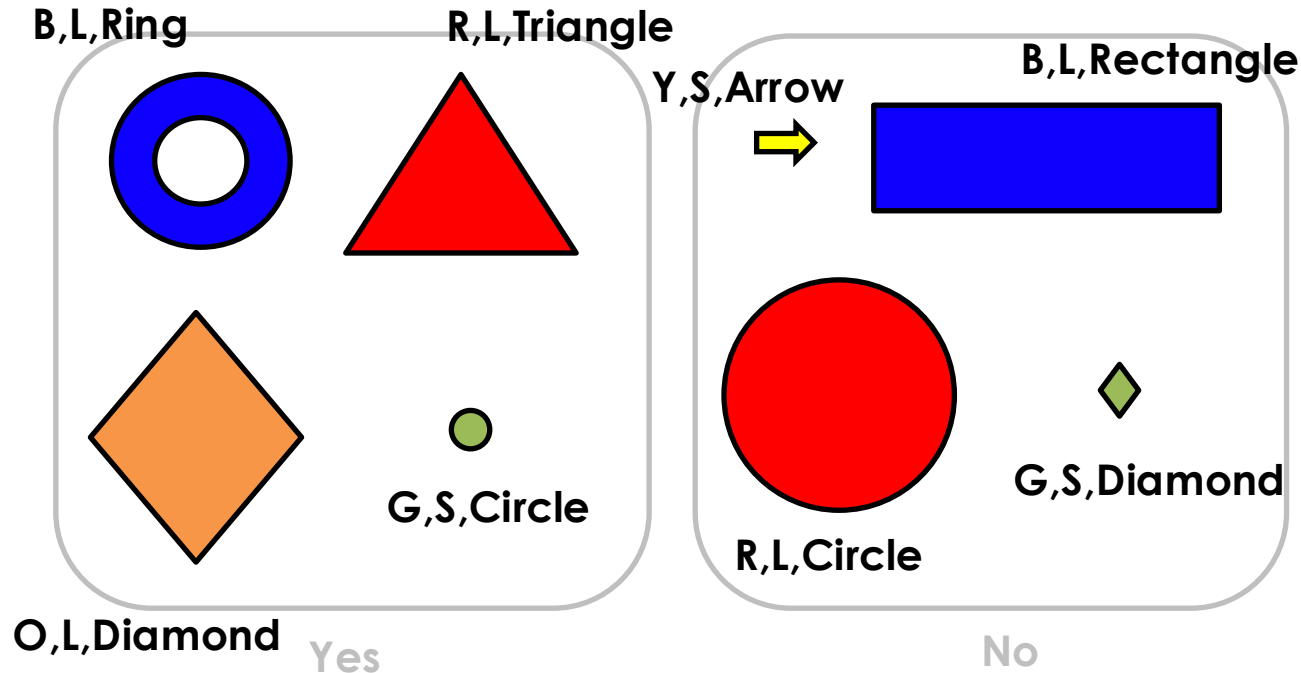


Step 1: Feature Extraction
Extract Attributes of Samples



Step 2: Sample Classification
Decide Label for a Sample

Walking Through A Toy Example: Classifying Tokens



Step 1: Feature Extraction
Color, Size, Shape

Walking Through A Toy Example: Token Classification

Feature Extraction

	Color	Size	Shape	Label
	Blue	Large	Ring	Yes
	Red	Large	Triangle	Yes
	Orange	Large	Diamond	Yes
	Green	Small	Circle	Yes
	Yellow	Small	Arrow	No
	Blue	Large	Rectangle	No
	Red	Large	Circle	No
	Green	Small	Diamond	No
	Yellow	Large	Triangle	?

Walking Through A Toy Example: Token Classification

Feature Extraction

	Color	Size	Shape	Label
	Blue	Large	Ring	Yes
	Red	Large	Triangle	Yes
	Orange	Large	Diamond	Yes
	Green	Small	Circle	Yes
	Yellow	Small	Arrow	No
	Blue	Large	Rectangle	No
	Red	Large	Circle	No
	Green	Small	Diamond	No

Walking Through A Toy Example: Token Classification

Feature Extraction

	Color	Size	Shape	Label
	Blue	Large	Ring	Yes
	Red	Large	Triangle	Yes
	Orange	Large	Diamond	Yes
	Green	Small	Circle	Yes
	Yellow	Small	Arrow	No
	Blue	Large	Rectangle	No
	Red	Large	Circle	No
	Green	Small	Diamond	No

Similarity

		0	1	0	1
		0	1	1	2
		0	1	0	1
		0	0	0	0
		1	0	0	1
		0	1	0	1
		0	1	0	1
		0	0	0	0

Walking Through A Toy Example: Token Classification

Similarity

		Color	Size	Shape	Total
		0	1	0	1
		0	1	1	2
		0	1	0	1
		0	0	0	0
		1	0	0	1
		0	1	0	1
		0	1	0	1
		0	0	0	0

Step 2: Sample Classification

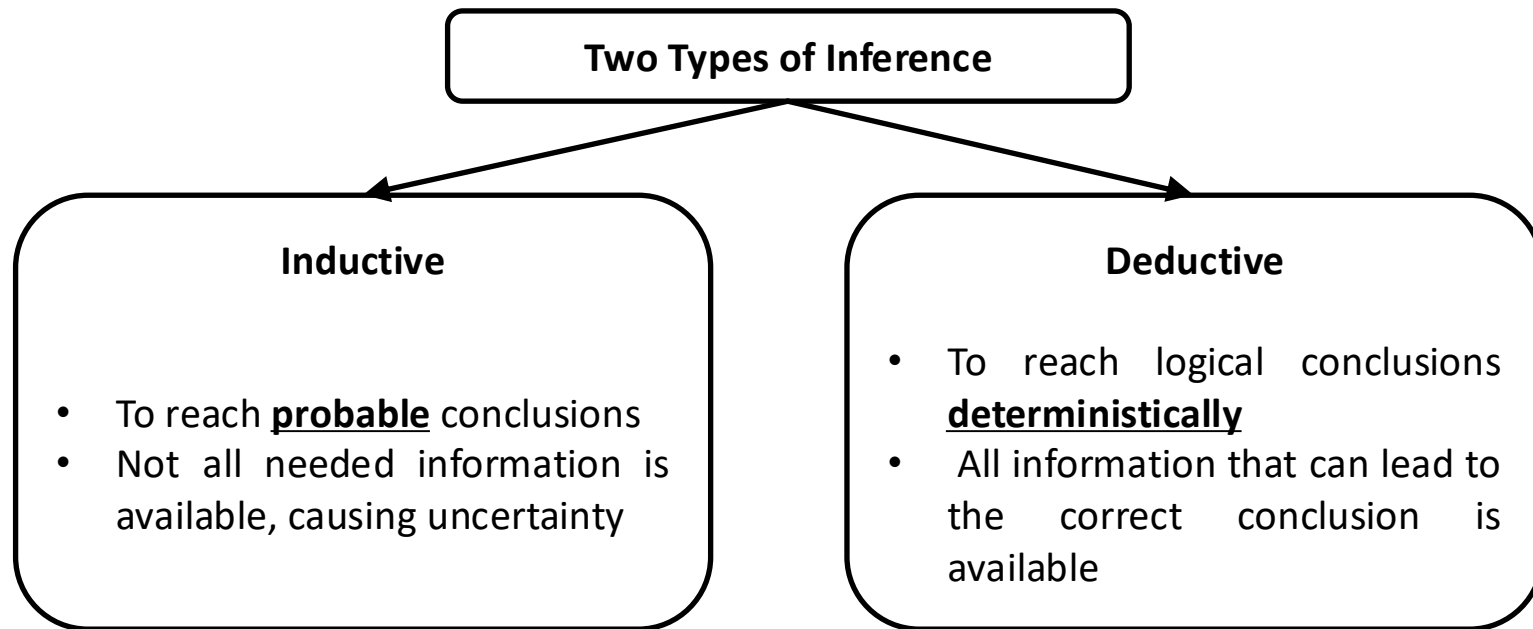
Nearest Neighbor Classifier:

- 1) Find the “nearest neighbor” of a sample in the feature space
- 2) Assign the label of the nearest neighbor to the sample

Nearest Neighbor Classifier is just one possible classifier. We may use other types of classifiers, given the extracted features.

Inductive vs. Deductive Reasoning

- Main Task of Machine Learning: to make inference



Probability and Statistics



Rule-based reasoning

NUS is in Singapore, Singapore is in Asia ->
NUS is in Asia

Inductive Reasoning

Note: humans use inductive reasoning all the time and not in a formal way like using probability/statistics.

B. C.

by Johnny hart



Ref: Gardener, Martin (March 1979). "MATHEMATICAL GAMES: On the fabric of inductive logic, and some probability paradoxes" (PDF). *Scientific American*. 234

Summary by Quick Quiz

Three Components in ML Definition

Task T, Performance P, Experience E

Three Types of in ML

Supervised Learning

Unsupervised Learning

Reinforcement Learning

Two Types of Supervised Learning

Classification, Regression

One Type of Unsupervised Learning

Clustering

Inductive and Deductive

Inductive: Probable

Deductive: Rule-based

Example of a Classifier Model

Nearest Neighbor Classifier

Practice Question

(Type of Question to Expect in Exams)

Which of the following statement is true?

- A. Nearest Neighbor Classifier is an example of unsupervised learning
- B. Nearest Neighbor Classifier is an example of deductive learning
- C. Nearest Neighbor Classifier is an example of feature selection
- D. None of the above is correct.

